

Vamsi Makke

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PROFESSIONAL SUMMARY

Software Engineer with **3+ years of experience** who owns the entire infrastructure at a **15,000+ MAU** startup, architecting microservices that handle **500+ daily video uploads** and real-time messaging for thousands of concurrent users. Previously built enterprise resource management systems at Cognizant serving **1,000+ daily users**. Core strengths in **distributed systems**, **API performance optimization**, and **event-driven architectures** using Java, Python, and cloud-native AWS services. Strong DSA fundamentals with **1,500+ LeetCode problems** solved (Top **4%** globally).

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, Bash

Backend: Spring Boot, FastAPI, Node.js, Express.js, REST APIs, Microservices, WebSockets

Frontend: React.js, HTML/CSS, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, Bedrock, CloudWatch), Docker, Kubernetes, GitHub Actions, CI/CD

Tools: Git, Apache Kafka, JUnit, Jest, Postman, Agile/Scrum

AI-Assisted Development: Claude, Cursor, Amazon Q

PROFESSIONAL EXPERIENCE

Software Engineer | TORQ Sports, Cincinnati, OH (Remote) Jun 2025 – Present

- Own the entire backend infrastructure for a sports social platform: **4 microservices** (API, Upload, Transcoding, Algorithm) across **Node.js** and **Python FastAPI**, **3 PostgreSQL databases (80+ tables)**, serving **15,000+ MAU**.
- Built end-to-end video pipeline: two-phase processing (sync MP4 remux → async **HLS** transcoding to 4 quality levels), plus **AWS Bedrock Claude Haiku** analysis that extracts frames and classifies sport type, content flags, and engagement scores for the feed algorithm.
- Designed feed ranking algorithm combining time-decay (6-day half-life), AI-derived quality metrics, and log-scaled engagement, incremental updates every 15 minutes, full re-ranking daily.
- Reduced **p95 API latency from 1s to 300ms** through PostgreSQL query optimization, composite indexing, and **Redis** caching, eliminating user-reported timeout complaints.
- Built roster management system: real-time messaging, polls with anonymous voting, event scheduling with push reminders - serving coaches, players, parents, scouts, and agents across **4 sports**.
- Implemented Trackman/Rapsodo data import for baseball analytics: CSV parsing with schema detection, pitch-by-pitch storage, and session summaries for coaches.
- Set up **CI/CD** with **GitHub Actions** and **Docker** for zero-downtime deployments, plus **CloudWatch** monitoring with custom metrics, cut incident detection from 30 min to under 5 min.
- Leveraged AI-assisted development tools (**Claude**, **Cursor**, **Amazon Q**) for rapid prototyping, debugging, and code reviews, accelerating feature delivery as a solo backend engineer.

Software Engineer | Cognizant Technology Solutions, Hyderabad, India Jan 2022 – Dec 2023

- Developed full-stack features for enterprise resource management platform using **Java**, **Spring Boot**, **React.js**, and **PostgreSQL**, serving **1,000+ daily active users** across 5 business units.
- Optimized critical database queries by implementing composite indexes and refactoring complex joins, reducing report generation time from **47 seconds to 8 seconds**, enabling real-time decision-making for resource managers.
- Implemented **Redis** caching layer for frequently accessed organizational data, reducing database load by **30%** and cutting infrastructure costs by eliminating need for database scaling.
- Developed resource booking REST APIs with comprehensive validation for concurrent booking conflicts and timezone handling across multiple regions.
- Increased unit test coverage from 40% to **85%** using **JUnit** and **Mockito**, reducing production defects by **60%** and enabling confident refactoring of legacy modules.
- Led containerization initiative using **Docker**, enabling consistent deployments and improving release frequency from bi-weekly to daily, accelerating feature delivery to stakeholders.

EDUCATION

University of Cincinnati, Cincinnati, OH
Master of Science in Information Technology | **GPA: 4.0 / 4.0**

Jan 2024 – May 2025

Vasireddy Venkatadri Institute of Technology, India
Bachelor of Technology in Computer Science and Engineering

Sep 2019 – Apr 2023

PROJECTS

Distributed E-Wallet System | *Java, Spring Boot, Apache Kafka, Redis, Docker*

- Architected microservices-based digital wallet (User, Wallet, Transaction, Notification services) with **Apache Kafka** for event-driven communication, choosing Kafka over RabbitMQ for its durability guarantees and replay capabilities essential for financial audit trails.
- Implemented **Saga pattern** with choreography-based coordination for distributed transactions, enabling automatic rollback across services when any step fails, critical for maintaining consistency without distributed locks.
- Built **Redis distributed locking** using Redlock algorithm to prevent double-spend scenarios, validated with **500 concurrent transactions** achieving zero data inconsistency. Designed for horizontal scaling to 10K+ TPS.

Real-Time Anomaly Detection Engine | *Python, FastAPI, AWS Bedrock, Pinecone*

- Built ML-powered log analysis system using **AWS Bedrock** (Titan embeddings) for semantic vectorization and **Pinecone** for similarity search across **1M+ log entries**. Chose vector similarity over regex/rule-based matching to catch semantically similar errors with different syntax.
- Designed ingestion pipeline with batched embedding generation (100 logs/batch) to optimize Bedrock API costs while maintaining near-real-time indexing.
- Achieved **50ms p95 latency** for anomaly detection queries through Pinecone's ANN indexing, reducing false positive alerts by **60%** compared to rule-based approaches — enabling oncall engineers to focus on real issues.

ACHIEVEMENTS

- **LeetCode**: Top **4%** globally with contest rating **1898**, solved **1,500+** problems in data structures and algorithms.
- **Hackathon**: 3rd place at Virtusa Neural Hack S6 National Hackathon among **10,000+** participants.